

Diffusion-Pyramid: One Shot Generation for Text-to-Image Diffusion Models

Yutong Wu (ytongwu) ytongwu@umich.edu

Qianhui Zheng (qianhui) qianhui@umich.edu

Ruikai Yang (ruikai) ruikai@umich.edu

Motivation

DreamBooth: Multiple images + Text prompt => Personalized generation



Input images



in the Acropolis



swimming



sleeping



in a doghouse



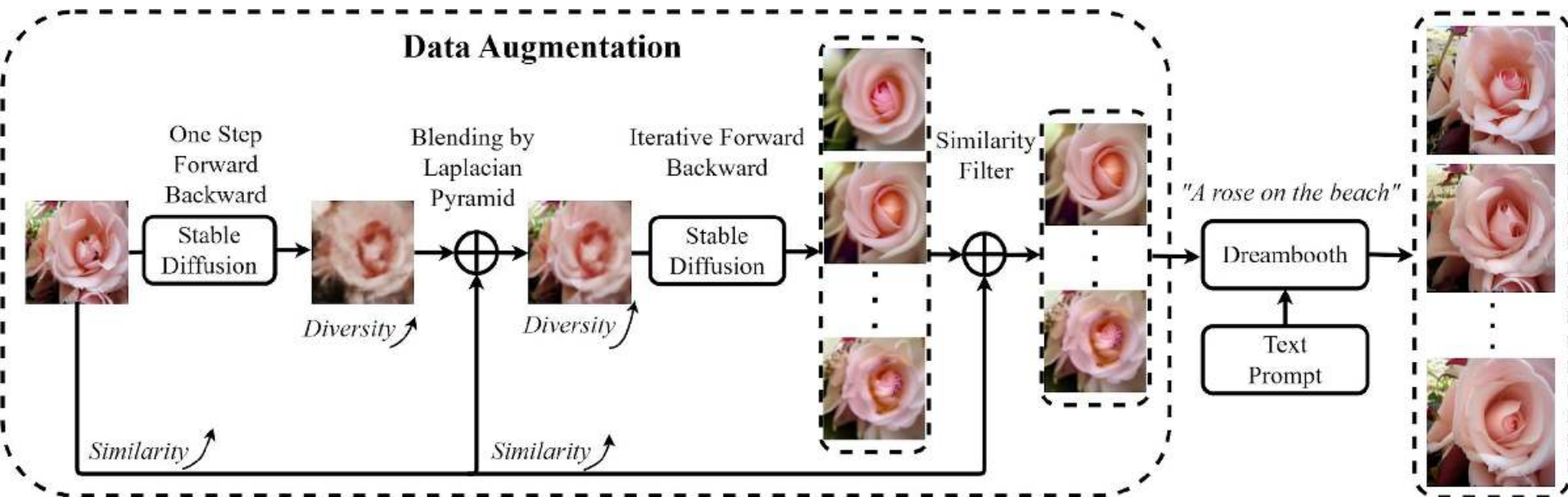
in a bucket



getting a haircut

Method

One Shot: Can we start from only one picture? Data Augmentation!



Result

“A rose on the beach”



Repetitive? Unreal?

“An insect on a daisy”



Improvement

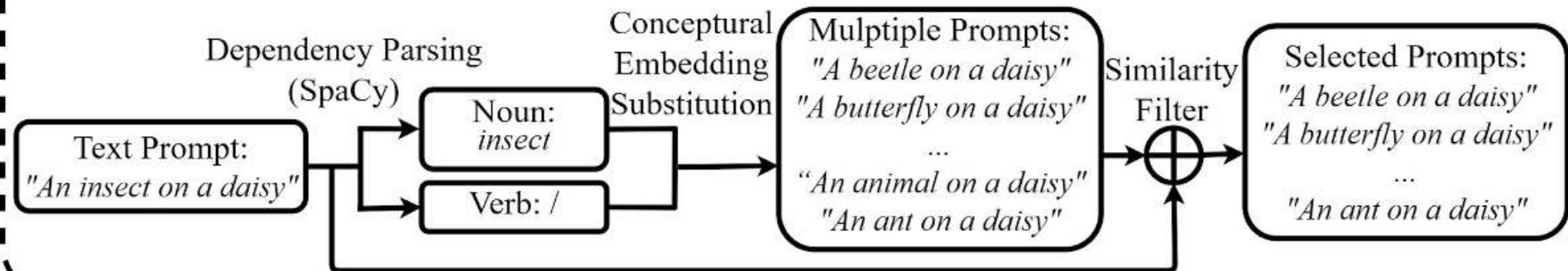
Repetitive? We want more prompts!

Unreal? We want more detailed prompts!

Then, can we **augment** our text prompts as well?



Text Augmentation



Text Augmentation

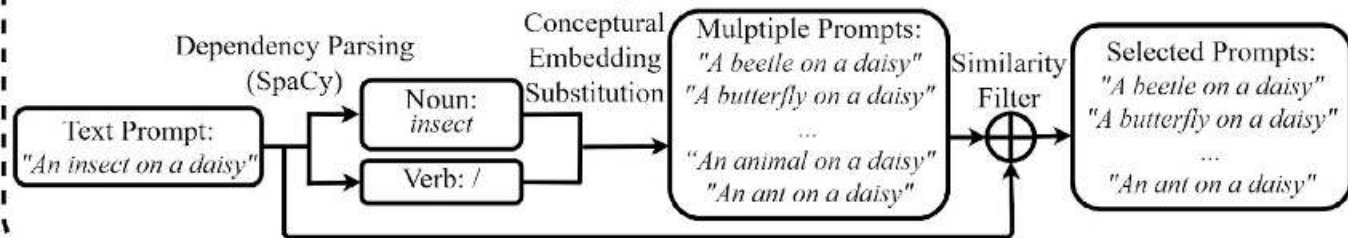
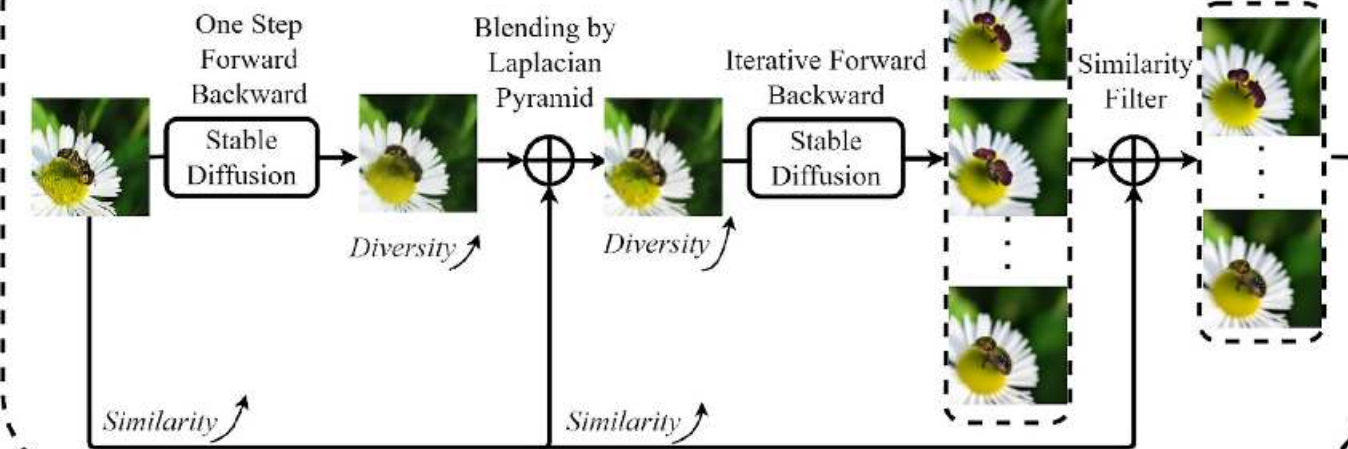


Image Augmentation



Dreambooth



"A beetle on a daisy"



"A butterfly on a daisy"



"An ant on a daisy"

Final Result

“An insect on a daisy”



Conclusion

Better Augmentations, Better Result ...

Methods	FID	Cos Similarity	SSIM	Cos Diversity
Cropping	351.32	0.9121	0.5877	0.8875
No augmentation	343.29	0.9158	0.5891	0.9036
Diffusion-Pyramid	340.45	0.9209	0.6097	0.8657

Reference

- [1] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. Dreambooth: Finetuning text-to-image diffusion models for subject-driven generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22500–22510, 2023.
- [2] Chun-Mei Feng, Kai Yu, Yong Liu, Salman Khan, and Wang-meng Zuo. Diverse data augmentation with diffusions for effective test-time prompt tuning. In *Proceedings of IEEE/CVF International Conference on Computer Vision*, pages 2704–2714, 2023.
- [3] Khawar Islam, Muhammad Zaigham Zaheer, Arif Mahmood, and Karthik Nandakumar. Diffusemix: Label-preserving data augmentation with diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27621–27630, 2024.

Input



Output ("*An insect on a daisy*")



Input



Input



Output ("*A bird on a daisy*")



Input



Output ("*Tulips with artistic genre*")



Input



Output ("*Sunflowers with artistic genre*")



Thank You!